

Trainings ▾

General Information

Cummins recommends the use of fully formulated antifreeze or coolant containing a precharge of Supplemental Coolant Additive (SCA). The antifreeze or coolant **must** meet the specifications outlined in The Maintenance Council (TMC) Recommended Practice (RP) 329 (ethylene glycol) or RP 330 (propylene glycol). The use of fully formulated antifreeze or coolant significantly simplifies cooling system maintenance.

Copies of TMC specifications can be obtained through Cummins, or by contacting:

The Maintenance Council
American Trucking Association
2200 Mill Road
Alexandria, VA 22314-5388
Phone (703) 833-1763
Fax (703) 836-6070

Fully-formulated antifreeze contains balanced amounts of antifreeze, SCA, and buffering compounds, but does **not** contain 50% percent water. Fully-formulated coolant contains balanced amounts of antifreeze, SCA, and buffering compounds already premixed 50/50 with deionized water.

The following pages give an explanation of water, antifreeze, and SCA's. They also explain how to test antifreeze and SCA levels.

This section also contains information on cooling system maintenance and a coolant treatment chart used to determine the correct SCA service filter.

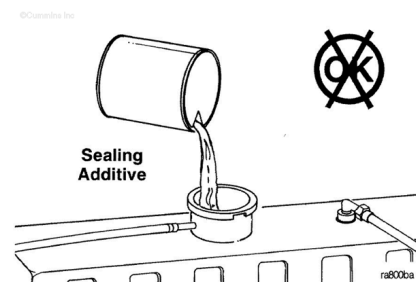
Alternative maintenance practices for cooling systems can be found in Cummins Coolant Requirements and Maintenance, Bulletin 3666132 ().

Cooling System Sealing Additives

Do **not** use sealing additives in the cooling systems. The use of sealing additives will:

- Build up in coolant low flow areas
- Clog coolant filters

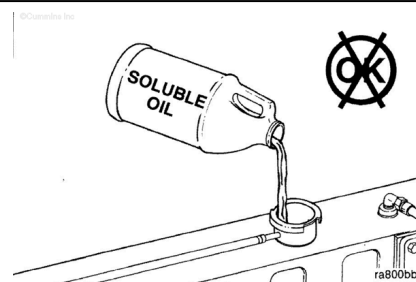
- Plug radiator and oil cooler.



Cooling System Soluble Oils

Do **not** use soluble oils in the cooling system. The use of soluble oils will:

- Allow cylinder liner pitting
- Corrode brass and copper
- Damage heat transfer surfaces
- Damage seals and hoses.



Fleetguard® DCA4 Service Filters and Liquid Precharge

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

DCA4 Service Filters		DCA (Fleetcool) Service Filters	
Part Number	SCA Units	Part Number	SCA Units
WF2070	2	WF2050	2
WF2071	4	WF2051	4
WF2072	6	WF2052	6
WF2073	8	WF2053	8
WF2074	12	Not Available	12
WF2075	15	WF2054	15
WF2076	23	WF2055	23
WF2077	(blank filter without SCAs)	WF2077	(blank filter without SCAs)

DCA4 Liquid		DCA (Fleetcool) Liquid			
Part Number	Size	SCA Units	Part Number	Size	SCA Units
DCA60L	0.47 liters [1 pt.]	5	DCA30L	0.47 liters [1 pt.]	5
DCA65L	1.89 liters [2 qt.]	20	DCA35L	1.89 liters [2 qt.]	20
DCA70L	3.78 liters [1 gal]	40	DCA40L	3.78 liters [1 gal]	40
DCA75L	18.9 liters [5 gal]	200	DCA45L	18.9 liters [5 gal]	200
DCA80L	208 liters [55 gal]	2200	DCA50L	208 liters [55 gal]	2200

Maintenance Intervals for Cooling Systems up to 76 Liters [20 Gallons]

Service Interval		System Size in Liters [Gallons]				
Kilometers	[Miles]	Hours	4 to 19 [1 to 5]	19 to 38 [6 to 10]	42 to 57 [11 to 15]	60 to 76 [16 to 20]
72001 to 80000	[45001 to 50000]	1126 to 1250	8	12	23	30
64001 to 72000	[40001 to 45000]	1001 to 1125	4	12	15	26
56001 to 64000	[35001 to 40000]	876 to 1000	4	8	12	23
48001 to 56000	[30001 to 35000]	751 to 875	4	6	12	20
40001 to 48000	[25001 to 30000]	626 to 750	4	6	10	18
32001 to 40000	[20001 to 25000]	501 to 625	2	6	8	15
24001 to 32000	[15001 to 20000]	376 to 500	2	4	6	12

**Maintenance Intervals for Cooling Systems up to 76
Liters [20 Gallons]**

Service Interval		System Size in Liters [Gallons]				
16001 to 24000	[10001 to 15000]	251 to 375	2	4	6	8
0 to 16000	[0 to 10000]	0 to 250	2	2	4	6
		Install service filter(s) and/or liquid containing number of SCA units listed above.				

Mainten ance Inter vals for Cooli ng Syste ms up to 1514 Liters [400 Gallo ns]										
Servi ce Inter val [Hour s]	Syste m Size in Liters [Gallo ns]									
	79 to 144 [21 to 30]	117 to 189 [31 to 50]	193 to 284 [51 to 75]	288 to 378 [76 to 100]	382 to 568 [101 to 150]	572 to 757 [151 to 200]	761 to 946 [201 to 250]	950 to 1135 [251 to 300]	1139 to 1325 [301 to 350]	1329 to 1574 [351 to 400]
751 to 1000	25	50	80	100	150	200	250	300	350	400
501 to 750	20	35	60	75	110	150	190	225	260	300
251 to 500	15	25	40	50	75	100	125	150	175	200
0 to 250	10	15	20	25	40	50	65	75	90	100

Instal l servi ce filter(s) and/ or liquid conta ining numb er of SCA units abov e.
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Notes:

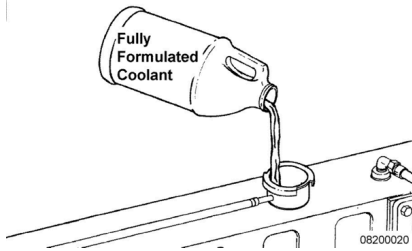
- A. Consult the vehicle equipment manufacturer's maintenance information for total cooling system capacity.
- B. When draining and replacing the coolant, **always** pre-charge the cooling system to a SCA level of 1.5 units per gallon. This concentration level **must not** be allowed to go below 1.2 units and **must** be controlled when the level is greater than 3 units. Action needed when the level goes below 1.2 is a filter and liquid pre-charge; from 1.2 to 3.0 units, filter **only**; above 3.0, test at every oil change until level falls to 3.0 or below.

When performing service which requires draining the cooling system, take special precautions to collect it in a clean container, seal it to prevent contamination, and save for reuse.

- A. Change coolant filters at each oil change to protect the cooling system. Consult the coolant capacity chart to determine the correct coolant filter for a given cooling system capacity and oil drain interval.

Supplemental Coolant Additive (SCA)

Fully-formulated products contain SCA's and are required to protect the cooling system from fouling, solder blooming, and general corrosion. The cooling filter is required to protect the coolant system from abrasive materials, debris, and precipitated coolant additives.



Supplemental coolant additives, or equivalent, are used to prevent liner pitting, corrosion, and scale deposits in the cooling system.

Use the correct Fleetguard® coolant filter to maintain the recommended SCA concentration in the system.

Maintain the correct concentration by changing the service filter at each oil drain interval.

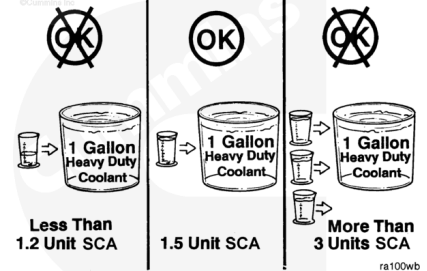
The correct filter is determined by the total cooling system capacity and oil drain interval. Refer to the Coolant Capacity Charts.



⚠ CAUTION ⚠

Insufficient concentration of the coolant additives will result in cylinder liner pitting, corrosion, solder blooming, and scale deposits which will lead to engine failure.

The SCA concentration **must not** fall below 1.2 units or exceed 3 units per gallon of cooling system capacity.



Use the correct Fleetguard® coolant filter to maintain the recommended SCA concentration in the system.

Maintain the correct concentration by changing the service coolant filter at each oil drain interval.

The correct filter is determined by the total cooling system capacity and oil drain interval.



Testing SCA Concentration Level CC-2602 Test Kit

Carefully follow the instructions to test the coolant and take the appropriate action recommended by the kit.

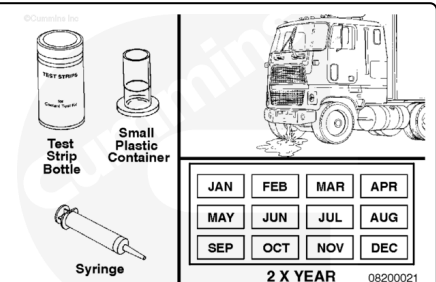
Precautions and Instructions for Proper Kit Use

- The coolant sample to be tested **must** be between 10 and 54°C [50 and 130°F]. If the sample is too cold or too hot, you will get incorrect results.
- To get the best color match results, compare test strip pads to the color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test does fall between two colors on the color chart, choose the lower numbered block. It is safer to underestimate your results than to overestimate.
- The test strips do have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect the life of the strips.
- Keep the cap tightly sealed on the test strip bottle except when removing a strip. Store away from direct sunlight and in an area where the temperature will generally stay below 32°C [90°F].
- Do **not** use the test strips after the expiration date stamped on the bottle.
- Discard the kit if any of the pads on the unused strips have turned light brown or pink.
- Use one strip at a time and take care **not** to touch any of the pads on the strip. Doing so will contaminate the pads and affect the test results.
- If the strip container is left uncapped for 24 hours, moisture in the air will render the strips useless, although no discoloration will be evident.
- **Only** use the color chart supplied with the kit.
- Clean and dry the sample cup and syringe after each use. This will prevent contaminating future samples.
- Following the correct test times is very important. Use a clock or stopwatch.

Test Intervals

Testing is recommended if the operator is **not** sure of cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

Testing is also recommended twice a year to monitor the SCA level. If the SCA level is above 3 units, test at subsequent oil drain intervals until the concentration is back under 3 units. When the concentration is back under 3 units, start installing the correct service filters at each drain interval.



If the concentration is below 1.2 units per gallon, replace the filter and precharge with liquid.



1.2 Units of SCA

1 Gallon Fully Formulated Coolant

OK

3 Units of SCA

1 Gallon Fully Formulated Coolant

OK


 25% 33% 40% 50% 60%
 +10 +5 0 -5 -10 -20 -30 -45 -60

S&A UNITS PER GALLON

SODIUM IODIDE WEIGHT LEVEL

	0.0	1.7	2.8	3.1	3.7	4.1	4.9	5.7
Row 6	0.0	1.7	2.3	2.7	3.1	3.5	4.3	5.1
Row 5	0.0	1.4	1.8	2.0	2.4	2.8	3.6	4.4
Row 4	0.0	1.2	1.5	1.7	2.1	2.5	3.3	4.1
Row 3	0.0	1.0	1.2	1.4	1.8	2.2	3.0	3.8
Row 2	0.0	0.6	0.9	1.1	1.5	1.9	2.7	3.5
Row 1	0.0	0.3	0.6	0.8	1.2	1.6	2.4	3.2

SODIUM NITRITE LEVEL

A B C D E F G H

Part Number	Description
CC2602 Coolant Test Kit	- Works with any SCA formulation (Call 1-800-521-4005) if you have this test kit and color chart does not show the number of units of SCA per gallon of coolant.
Probablizer:	
3318169S Plug	- Installs on the engine for easy coolant sampling
3318168S Cap	- Use with Monitor C bottle to sample coolant
CC2700 Monitor C	- Lab analysis of coolant sample

To get answers to any questions about cooling system maintenance, contact your Local Cummins Authorized Repair Location.

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**Cummins: 1-800-DIESELS
1-800-521-4005**

**1-800-22-FILTERS
1 - 800 - 223 - 4583**

00200003

Coolant Replacement Requirements

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain and flush the cooling system after 6,000 hours, or 2 years of service. Refill with either new fully-formulated coolant or a 50/50 mixture of good quality water and fully-formulated antifreeze, and install the correct service coolant filter.

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